BILAL AHAMAD

Systems Validation Architect

Sunnyvale, CA · contact@bilalahamad.com · bilalahamad.com · linkedin.com/in/bilalahamad · github.com/bilalahamad0 · Phone on request

SUMMARY

Systems validation architect with 18+ years building embedded validation ecosystems for safety-critical systems — firmware, IoT, wearables, AI video, and autonomous driving. Takes products from NPI bring-up to enterprise release through test automation, SIL environments, and test strategy.

CORE SKILLS

Languages & Automation: Python, Shell, C/C++, JavaScript · Pytest, Selenium, Appium

Embedded, OS & Protocols: Embedded Linux, RTOS, QNX, Android/AAOS, ARM, ESP32, Raspberry Pi · BLE, LTE/Wi-Fi, WFA certification, RTSP/RTP, TLS, UART/SPI/I2C, CAN FD, Automotive Ethernet, V2X, Wireshark/PCAP

Infrastructure & Observability: CI/CD (GitLab, GitHub Actions, Buildkite, Jenkins) · SIL (Docker/QEMU), HIL benches · Splunk, Grafana, Datadog

EXPERIENCE

System Architect & Technical QA Lead — Stealth Mode — Sep 2025 - Oct 2026 · Remote / Sunnyvale, CA

- Built a LiDAR/Radar bring-up platform in Python with Docker/QEMU SIL (virtual-ECU) environments and a pre-validated ARM/ESP32 compatibility layer, cutting bring-up cycles by 70%.

Senior Firmware Quality Lead — Samsara Inc — Dec 2023 - Jul 2025 · Remote / San Francisco, CA

- Cut the regression cycle from two weeks to five days and post-launch firmware escapes by 30% across the Dash Cam product line.

Senior Automation Architect — Cruise LLC — Oct 2022 - Jun 2023 · San Francisco, CA

- Engineered a Python framework stress-testing autonomous driving compute workloads across CPU/GPU integration, AI/ML performance, V2X and PCAP replay, reaching 75% automation coverage.

Test Lead & Senior System Test Engineer — Rivian Automotive LLC — Jun 2021 - Sep 2022 · Palo Alto, CA

- Reduced production-blocking navigation and connectivity issues by 40% before first customer ship, coordinating 3+ squads on a bi-weekly release train.

Senior Product Quality Lead & QAE II — Amazon Lab126 — Jun 2018 - Jun 2021 · Sunnyvale, CA

- Engineered a reusable automation library of AVS test primitives, reducing manual testing cost by $3.0M.

Senior Test Engineer — Tech Mahindra / Google Inc — Jan 2016 - Jun 2018 · Mountain View, CA

- Designed and built a 3DOF robotic motion-stimulus fixture (Arduino, servo motors) automating motion-fidelity, IMU sensor-fusion and spatial-tracking validation, cutting manual execution hours by 80%.

Earlier (2008–2016): Cognizant Technology / Cisco · Wistron Mobile Solutions Corp · L&T Infotech / Motorola Mobility · Luminous Infoways

SELECTED PROJECTS

US Live Layoff Monitoring Dashboard · Adhan Caster — Cross-Browser Extension · Smart-Home IoT Media Caster — shipped, open-source; details at bilalahamad.com/projects

CERTIFICATIONS & EDUCATION

Certifications: Google AI Professional (2026) · Google Project Management Professional (2026) · ISTQB Certified Tester Foundation Level (CTFL)

B.Tech, Electronics & Telecommunications Engineering — Biju Patnaik University of Technology, 2004 – 2008